

OAKPOINT, MB, Canada

WMO: 711440

Lat: 50.500N

Lon: 98.037W

Elev: 819

StdP: 14.27

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.5	-20.4	-30.5	1.0	-24.3	-26.2	1.3	-20.1	29.8	4.8	26.7	4.8	6.6	350	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	81.4	71.4	78.7	69.6	76.5	68.1	74.1	79.5	71.9	77.1	69.8	75.2	9.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.3	123.4	78.3	70.0	113.6	76.2	67.7	105.0	73.9	38.4	79.7	36.3	77.1	34.4	75.1	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.1	24.7	21.6	-30.6	86.5	5.3	3.0	-34.5	88.7	-37.6	90.4	-40.6	92.1	-44.5	94.2		
			-30.3	77.2	5.4	2.4	-34.2	78.9	-37.3	80.3	-40.3	81.7	-44.3	83.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	1.8	5.1	18.1	34.8	49.2	61.8	67.2	65.1	55.6	41.4	24.4	9.0	(6)
	DBStd	25.23	13.88	13.64	13.24	10.60	9.04	6.82	5.65	5.97	7.76	7.96	10.96	12.80	(7)
	HDD50	6693	1493	1258	989	463	127	4	0	33	286	769	1271		(8)
	HDD65	10723	1958	1678	1454	906	492	137	44	73	293	733	1219	1736	(9)
	CDD50	1690	0	0	0	8	102	359	532	469	202	18	0	0	(10)
	CDD65	244	0	0	0	0	3	42	110	77	12	0	0	0	(11)
	CDH74	1120	0	0	0	0	15	169	523	376	37	0	0	0	(12)
	CDH80	138	0	0	0	0	1	19	62	53	3	0	0	0	(13)
	WSAvg	10.4	10.0	9.9	10.0	9.8	10.1	9.9	10.1	10.2	10.8	11.8	11.4	10.3	(14)
	PrecAvg	20.4	0.8	0.7	1.1	1.3	2.6	3.5	2.7	2.2	1.9	1.5	1.1	1.1	(15)
(16) Precipitation	PrecMax	28.0	2.2	1.7	2.6	3.0	5.7	6.1	4.8	5.0	4.1	3.5	4.6	2.0	(16)
	PrecMin	15.7	0.3	0.0	0.2	0.4	0.8	1.2	0.8	0.6	0.7	0.2	0.1	0.3	(17)
	PrecStd	3.0	0.4	0.5	0.7	0.7	1.4	1.3	1.1	1.2	0.9	0.9	1.0	0.5	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	34.6	36.8	45.5	64.8	75.8	82.6	84.9	85.2	78.0	65.0	49.6	35.7		(19)
	0.4% MCWB	33.5	35.1	40.8	50.6	60.4	72.1	74.9	72.9	67.1	56.4	45.4	33.6		(20)
	2% DB	30.5	33.6	39.5	58.4	70.5	78.3	81.6	80.9	73.8	59.9	44.6	31.9		(21)
	2% MCWB	29.5	31.8	36.4	48.4	58.6	67.7	72.6	70.2	65.1	53.8	41.3	30.8		(22)
	5% DB	26.5	30.8	36.8	53.2	66.1	75.5	79.1	78.0	70.5	55.9	41.1	29.3		(23)
	5% MCWB	25.7	29.3	34.8	45.0	56.6	66.7	71.0	68.6	63.3	51.0	38.7	28.4		(24)
	10% DB	22.7	25.7	34.7	49.0	62.6	72.8	76.8	75.5	67.4	52.9	38.1	26.7		(25)
	10% MCWB	21.9	24.6	33.2	43.1	54.6	65.1	69.6	67.1	61.4	48.8	36.1	25.7		(26)
	0.4% WB	33.8	35.4	41.6	52.7	64.7	75.3	81.1	76.7	69.7	58.4	45.8	33.9		(27)
	0.4% MCDB	34.5	36.9	45.2	62.0	71.4	80.6	82.8	82.8	75.1	64.0	48.7	35.2		(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	29.8	32.2	37.0	49.3	61.0	70.9	74.6	73.3	66.8	54.3	41.6	31.2		(29)
	2% MCDB	30.5	33.7	38.9	56.2	67.8	76.0	79.7	79.2	72.0	58.7	43.9	32.0		(30)
	5% WB	26.1	29.5	35.1	46.5	58.4	68.3	72.7	70.9	64.6	51.7	38.8	28.7		(31)
	5% MCDB	26.8	31.1	36.7	52.6	64.3	73.5	77.9	76.4	69.7	55.4	40.8	29.5		(32)
	10% WB	22.1	24.9	33.4	43.1	55.9	66.0	70.7	68.7	62.3	49.0	36.1	26.2		(33)
	10% MCDB	22.9	26.0	34.7	49.0	61.6	71.6	76.0	74.1	67.1	52.7	37.9	27.0		(34)
(35) Mean Daily Temperature Range	MDBR	14.9	16.7	16.9	16.5	16.8	15.3	15.0	16.2	15.0	12.1	10.8	13.1		(35)
	5% DB MCDBR	16.0	13.5	14.1	22.1	21.2	17.8	16.1	18.5	18.7	16.9	11.7	13.1		(36)
	5% DB MCWBR	15.6	12.3	11.7	13.4	12.6	11.5	10.9	11.8	12.2	11.5	9.2	12.6		(37)
	5% WB MCDBR	15.6	13.0	13.8	21.4	17.8	15.9	15.3	16.9	16.9	15.7	11.0	13.0		(38)
	5% WB MCWBR	15.2	11.9	11.7	13.6	11.9	11.5	11.5	12.0	12.3	11.5	8.9	12.6		(39)
(40) Clear-Sky Solar Irradiance	taub	0.253	0.282	0.306	0.355	0.371	0.379	0.392	0.379	0.346	0.300	0.265	0.242		(40)
	taud	2.268	2.202	2.291	2.342	2.368	2.381	2.373	2.407	2.482	2.561	2.422	2.295		(41)
	Ebn at Noon	250	270	285	281	281	279	272	271	270	266	248	238		(42)
	Edh at Noon	21	30	33	36	36	36	36	33	28	21	18	18		(43)
(44) All-Sky Solar Radiation	RadAvg	414	791	1218	1611	1746	1848	1945	1615	1155	676	365	296		(44)
	RadStd	37	53	91	110	155	137	84	119	95	88	52	30		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	-1.03	-1.42	N/A	N/A	N/A	N/A

Nomenclature: See separate page